

Notes: Chiang and Knight (RES, 2011)
Media Bias and Influence: Evidence from Newspaper Endorsements

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1 Big picture

- **Question.** Do newspaper endorsements influence voting, and do voters discount endorsements that come from ideologically biased newspapers? The paper asks whether media bias affects the *influence* of political information, not only the content of media messages.
- **Setting.** The empirical setting is newspaper endorsements in the 2000 and 2004 U.S. presidential elections. The paper combines daily survey data on voting intentions and newspaper readership with the exact timing and candidate choice of newspaper endorsements.
- **Core theoretical idea.** Voters are uncertain about candidate quality. Newspapers receive better information but may be ideologically biased. Rational voters therefore learn more from endorsements that are more surprising relative to the newspaper's ideological position.
- **Main prediction.** A Democratic endorsement from a right-leaning newspaper is more credible than a Democratic endorsement from a left-leaning newspaper. Likewise, a Republican endorsement from a left-leaning newspaper is more credible than a Republican endorsement from a right-leaning newspaper.
- **Main empirical result.** Endorsements are influential: after a newspaper endorses a candidate, its readers become more likely to support the endorsed candidate.
- **Main credibility result.** The effect is much larger for high-credibility, surprising endorsements. Endorsements from very biased sources have little influence.
- **Voter heterogeneity.** Endorsements are more influential among moderate voters and voters more likely to be exposed to the endorsement, especially frequent newspaper readers. Some groups, such as born-again Christians, appear less responsive after accounting for credibility.
- **Economic takeaway.** The paper supports a sophisticated-voter interpretation. Voters use newspapers as information sources, but they do not treat all endorsements equally. They discount endorsements that predictably follow the newspaper's ideological bias.

2 Incremental contribution of the paper

- **From measuring media bias to measuring media influence.** Earlier work measures media slant or studies media exposure. This paper directly links bias to persuasive influence by asking whether biased and unbiased endorsements move voter intentions differently.
- **A credibility-based theory of persuasion.** The paper builds a simple Bayesian learning model in which the same endorsement can have different effects depending on the newspaper's editorial position. This connects media bias to rational updating.
- **Use of precise endorsement timing.** Instead of comparing readers and non-readers cross-sectionally, the paper uses the date of each endorsement and daily survey responses. This allows a before-versus-after comparison among readers of the same newspaper.
- **Endorsement credibility as a generated empirical object.** The paper estimates newspaper ideology in a first-stage endorsement model using reader preferences and ownership characteristics. It then transforms predicted ideology into Mills-ratio-style credibility measures.

- **Distinguishing surprising from unsurprising endorsements.** The key empirical contribution is not only showing that endorsements matter. It shows that *surprising* endorsements matter more, which is the core test of rational filtering of media bias.
- **Evidence against naive persuasion.** If voters naively followed newspapers, endorsements from biased newspapers should be just as influential. The results instead suggest that voters discount predictable endorsements.
- **Heterogeneous influence.** The paper goes beyond the average effect and shows that exposure and ideology matter. Credible endorsements have larger effects among voters who are more likely to read the editorial and among moderate voters.
- **Policy and political economy contribution.** The findings imply that media bias does not mechanically translate into electoral influence. The effect of biased media depends on whether audiences understand and filter the bias.

3 Data and measurement

3.1 Survey data on voters

- The voter data come from the National Annenberg Election Surveys for the 2000 and 2004 presidential elections.
- The surveys use a rolling cross-section design, polling voters daily in the months before the election.
- The key voter variables are interview date, voting intention or post-election vote, newspaper read most often, demographic characteristics, newspaper-reading frequency, party identification, and ideology.
- The main voting outcome is an indicator equal to one if the respondent intends to vote for the Democratic candidate.
- The paper excludes early voters because the timing of the vote relative to the endorsement is not observed precisely.

Interpretation: The daily survey structure is central. It lets the authors compare readers of a given newspaper immediately before and after that newspaper’s endorsement, while controlling for newspaper and date fixed effects.

3.2 Newspaper endorsement data

- The endorsement data record which candidate each newspaper endorsed and the exact date on which the endorsement appeared.
- Endorsement information is compiled from Democracy in Action and newspaper archives such as Lexis-Nexis, Factiva, and the Associated Press.
- Figure 1 shows substantial variation in endorsement timing. Many endorsements appear shortly before election day, but some occur earlier.
- The baseline sample excludes newspapers with fewer than ten survey readers and newspapers that did not make endorsements.

- The final sample contains 166 endorsing newspapers in 2000, 212 in 2004, and 32,014 individual respondents.

Interpretation: Exact timing is what makes the design credible. Without timing, endorsement effects would be confounded by readers choosing newspapers that already match their politics.

3.3 Measuring newspaper ideology and credibility

- Newspaper ideology is not directly observed. The paper estimates it using a first-stage model of endorsement decisions.
- The first-stage covariates include pre-endorsement reader preferences and newspaper ownership group indicators.
- Reader preference is measured as the fraction of pre-endorsement readers who support the Democratic candidate.
- Ownership group indicators proxy for supply-side editorial preferences.
- The first-stage predicted probability of endorsing the Democrat is transformed into credibility measures. A Democratic endorsement is more credible when the predicted probability of endorsing the Democrat is low. A Republican endorsement is more credible when the predicted probability of endorsing the Democrat is high.

Interpretation: Credibility is an inverse-surprise object. When a newspaper endorses the candidate it was already expected to favor, the endorsement conveys less information about candidate quality.

3.4 Outcome and control variables

- The dependent variable in most regressions is an indicator for intending to vote for the Democratic candidate.
- Individual controls include education, gender, race, age, age squared, born-again Christian status, religious attendance, and income categories.
- The baseline second-stage regressions include newspaper fixed effects and date fixed effects.
- Richer specifications add party and ideology controls, newspaper-specific trends, ideology-specific trends, and clustering at the newspaper level.

Interpretation: Newspaper fixed effects absorb stable differences across newspaper readerships. Date fixed effects absorb national campaign shocks. The remaining variation comes from whether a reader is surveyed before or after his or her newspaper publishes its endorsement.

4 Models and empirical specifications

4.1 Voter preferences

The model has two candidates, Democrat D and Republican R . Candidate ideology is $i_D < i_R$, voter ideology is i_v , and candidate quality is q_c . Voter v 's payoff from candidate c is

$$U_{vc} = q_c - \frac{\omega}{2}(i_v - i_c)^2. \quad (1)$$

Define relative quality as $q = q_D - q_R$. Before observing an endorsement, priors over relative quality have mean zero. A voter supports the Democrat when

$$i_v < \frac{i_D + i_R}{2}. \quad (2)$$

After observing newspaper n 's endorsement e_n , the voter supports the Democrat when

$$i_v < \frac{i_D + i_R}{2} + \frac{E(q | e_n)}{\omega(i_R - i_D)}. \quad (3)$$

Interpretation: An endorsement shifts the ideological cutoff. If it raises expected relative quality of the Democrat, the Democratic support cutoff moves to the right; if it lowers expected Democratic quality, the cutoff moves left.

4.2 Newspaper endorsement rule

Newspaper n receives a signal of relative candidate quality:

$$d_n = q + \varepsilon_n. \quad (4)$$

The newspaper has editorial threshold p_n and endorses the Democrat if the normalized signal exceeds that threshold:

$$e_n = 1 \left[\frac{d_n}{\sqrt{\sigma_q^2 + \sigma_\varepsilon^2}} > p_n \right]. \quad (5)$$

- A larger p_n means the newspaper is more right-leaning because it requires stronger evidence to endorse the Democrat.
- A smaller p_n means the newspaper is more left-leaning because it has a lower threshold for endorsing the Democrat.

Interpretation: The same endorsement has different meaning depending on p_n . A Democratic endorsement by a right-leaning paper must clear a high threshold, so it is informative.

4.3 Credibility and Bayesian updating

For a Democratic endorsement, voters update according to

$$E(q | e_n = 1) = \beta \lambda_D(p_n), \quad (6)$$

where

$$\lambda_D(p_n) = \frac{\phi(p_n)}{1 - \Phi(p_n)}. \quad (7)$$

For a Republican endorsement,

$$E(q | e_n = 0) = -\beta \lambda_R(p_n), \quad (8)$$

where

$$\lambda_R(p_n) = \frac{\phi(p_n)}{\Phi(p_n)}. \quad (9)$$

- $\lambda_D(p_n)$ is increasing in p_n : a Democratic endorsement is more credible when it comes from a right-leaning newspaper.

- $\lambda_R(p_n)$ is decreasing in p_n : a Republican endorsement is more credible when it comes from a left-leaning newspaper.
- β is the updating coefficient; it is larger when voters are initially more uncertain and smaller when newspaper signals are noisier.

Interpretation: The credibility measure is a formal expression of surprise. Voters learn more when a newspaper endorses against its own bias.

4.4 First-stage endorsement equation

Newspaper editorial position is parameterized as

$$p_n = \gamma Z_n, \quad (10)$$

where Z_n includes reader preferences and ownership-group variables. The first-stage endorsement equation is

$$\Pr(\text{endorse } D_n) = 1 - \Phi(\gamma Z_n). \quad (11)$$

Interpretation: This first stage estimates how likely a newspaper is to endorse the Democrat, given its readership and ownership. The predicted probability is then used to define endorsement credibility.

4.5 Second-stage voting equation

The main second-stage regression is

$$\Pr(\text{vote } D_{vnt}) = \beta \text{After}_{nt} [e_n \lambda_D(\gamma Z_n) - (1 - e_n) \lambda_R(\gamma Z_n)] - \theta X_v - \eta_t - \eta_n. \quad (12)$$

- After_{nt} equals one after newspaper n has published its endorsement by date t .
- e_n equals one if the endorsement is for the Democrat and zero if for the Republican.
- X_v includes voter-level controls.
- η_t are date fixed effects and η_n are newspaper fixed effects.
- The coefficient β is the key effect of a credibility-weighted endorsement.

Interpretation: The coefficient asks whether more credible endorsements produce larger shifts in intended vote among readers after the endorsement appears.

5 Identification and empirical design

5.1 What the design identifies

- The design identifies the association between endorsement publication and voting intentions among readers, conditional on newspaper fixed effects, date fixed effects, and voter controls.
- Because endorsement timing varies by newspaper and because voters are surveyed daily, the paper compares similar readers before and after endorsement publication.
- The key causal interpretation is strongest for the early campaign period when endorsements represent media information rather than reactions to unobserved local events.

Interpretation: The paper is not a randomized field experiment. Its strength is the combination of precise timing, fixed effects, and a theory-driven credibility measure.

5.2 Selection concerns

- Readers may select newspapers that match their ideology. Newspaper fixed effects help address this because the analysis compares readers of the same newspaper over time.
- Newspapers may endorse high-quality candidates who are also gaining support for other reasons. Date fixed effects absorb national swings, but not all local shocks.
- Endorsement timing may respond to campaign events. The paper examines timing persistence across election cycles to address this concern.
- Reader preferences are used in the first stage, which could create finite-sample measurement issues. The paper uses Monte Carlo checks and alternative historical endorsement measures.

Interpretation: The main remaining threat is not stable ideological sorting, but time-varying local shocks that affect both endorsement timing and voter preferences.

5.3 Robustness strategy

- The paper adds newspaper-specific trends and ideology-specific trends.
- It allows credibility to interact with date fixed effects.
- It focuses on moderates and excludes ideological extremes.
- It tests alternative credibility measures based on linear residuals and historical endorsement patterns.
- It adds party and ideology controls.
- It uses standard errors clustered at the newspaper level as a robustness check.

Interpretation: The robustness checks are designed to show that the main result is not an artifact of trend differences, timing, first-stage credibility construction, or standard-error assumptions.

6 Tables and figures

6.1 Figure 1 and Table 1: Timing and sample summary

Read:

- Figure 1 shows the distribution of endorsement dates in 2000 and 2004.
- Many endorsements occur in the final two weeks before election day, especially on weekends, but there is enough timing variation for before-versus-after comparisons.
- Table 1 reports the main survey sample: 32,014 observations.
- The mean Democratic vote intention is 0.544; 59.3% of respondents read a newspaper that endorsed the Democratic candidate.
- Roughly half of respondents are low-frequency newspaper readers, meaning they read the paper fewer than seven days per week.

Economic message: The timing variation makes the empirical strategy possible, while the summary statistics show that the paper studies a large daily survey sample of actual newspaper readers.

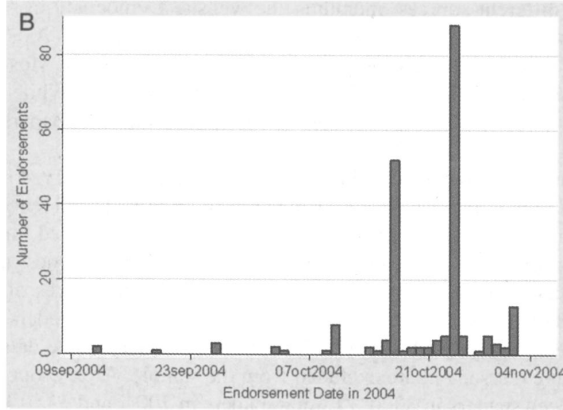
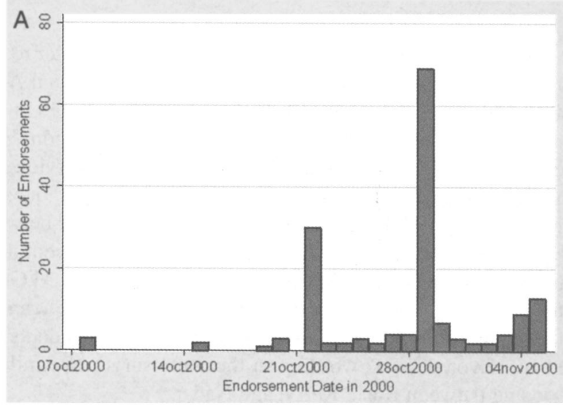


FIGURE 1
Dates of newspaper endorsements in 2000 and 2004

(a) Figure 1: Dates of newspaper endorsements

TABLE 1
Summary statistics

Variable	Observations	Mean	Standard deviation	Minimum	Maximum
Intend to vote for Democratic candidate	32,014	0.544	0.498	0	1
Have a high school degree, no college	32,014	0.217	0.412	0	1
Have some college or higher	32,014	0.741	0.438	0	1
Male	32,014	0.473	0.499	0	1
Black	32,014	0.093	0.291	0	1
Age	32,014	47.225	15.973	18	97
Born-again Christian	32,014	0.312	0.463	0	1
Attend religious services	32,014	0.388	0.487	0	1
Low-frequency readers	32,014	0.514	0.500	0	1
Read newspaper with democratic endorsement	32,014	0.593	0.491	0	1

(b) Table 1: Summary statistics

6.2 Tables 2 and 3: First-stage ideology and baseline influence

Read:

- Table 2 estimates the first-stage endorsement equation. Reader preference strongly predicts whether a newspaper endorses the Democrat.
- Some ownership groups are significantly more likely to endorse the Democratic candidate.
- The 2004 indicator is positive, implying newspapers were more likely to endorse the Democrat in 2004 than in 2000, conditional on the included covariates.
- Table 3 is the main second-stage evidence. In column I, the coefficient on *After* $\times$ *Credibility* is positive and statistically significant.
- Column II shows that a simple unweighted endorsement measure is small and insignificant.
- Column III includes both the credibility-weighted measure and the simple endorsement dummy; the credibility measure remains positive and significant.

Economic message: The paper’s central result is that endorsements matter because they convey credible information, not simply because newspapers tell readers which candidate to support.

TABLE 2
First stage—newspaper ideology

Dependent variable: endorse Democratic candidate	
Readers preference	4.073*** (0.673)
Group owner effect ¹	
Advance Publications Inc.	0.269 (0.280)
Cox Newspapers	1.096** (0.422)
E W Scripps Co.	−0.164 (0.526)
Gannett Co. Inc.	0.910*** (0.238)
Hearst Newspapers	0.463 (0.334)
Knight Ridder	1.043*** (0.324)
Lee Enterprises Inc.	−0.135 (0.470)
McClatchy Newspapers	1.634*** (0.381)
New York Times Co.	0.870** (0.395)
Year 2004	0.322** (0.155)
Constant	−2.500*** (0.377)

Observations: 378 Sample: Newspapers made endorsements in 2000 or 2004²
Notes: Standard errors in parentheses. * denotes 90% significance, ** denotes 95% significance, and *** denotes 99% significance. ¹ Default category: newspapers not owned by group owners. Companies own more than 10 newspapers in the sample are defined as group owner of newspapers. ² Newspapers with the same name in different years are treated as different newspapers.

TABLE 3
Second Stage: effect of newspaper endorsements on vote intention

Dependent variable: 1 if intend to vote for the Democrat			
	I	II	III
After×Credibility	0.029** (0.013)		0.055** (0.026)
After×Endorsement		0.011 (0.008)	−0.020 (0.017)
High school	−0.047*** (0.016)	−0.047*** (0.015)	−0.047*** (0.016)
College	−0.013 (0.016)	−0.013 (0.016)	−0.013 (0.016)
Male	−0.088*** (0.006)	−0.087*** (0.006)	−0.088*** (0.006)
Black	0.440*** (0.009)	0.440*** (0.008)	0.440*** (0.009)
Age	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
Age squared	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Born-again Christian	−0.150*** (0.007)	−0.150*** (0.007)	−0.150*** (0.007)
Attend religious activities	−0.123*** (0.006)	−0.123*** (0.006)	−0.123*** (0.006)
Constant	0.740*** (0.183)	0.740*** (0.189)	0.741*** (0.183)
Income categories	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Observations	32,014	32,014	32,014

Notes: Standard errors in parentheses. * denotes 90% significance, ** denotes 95% significance, and *** denotes 99% significance.

(a) Table 2: First stage - newspaper ideology

(b) Table 3: Second stage - endorsement effects

6.3 Table 4 and Figure 2: Top newspapers and average event pattern

Read:

- Table 4 illustrates credibility and implied influence for major newspapers in 2000.
- The New York Times is predicted to endorse Gore with high probability, so its Gore endorsement has low implied influence.
- The Denver Post and Chicago Sun Times have more surprising endorsements, and their implied influence is larger.
- Figure 2 plots Democratic support around endorsement dates for readers of Democrat-endorsing and Republican-endorsing newspapers.
- The average gap widens somewhat after endorsement publication, but the visual effect is modest when all endorsements are pooled.

Economic message: Average endorsement effects are hard to see unless credibility is considered. Table 4 shows why: predictable endorsements carry little information, while surprising endorsements are more powerful.

TABLE 4
Influence of top 20 newspapers in 2000[†]

Newspaper	Reader support for Gore (%)	Group owner [‡]	Probability of endorsing Gore (%)	Actual endorsement	Implied influence (%)
<i>New York Times</i>	75	New York Times	90	Gore	0.50
<i>Washington Post</i>	64	—	54	Gore	2.10
<i>New York Daily News</i>	67	—	58	Gore	1.90
<i>Chicago Tribune</i>	53	—	36	Bush	-1.70
<i>Newsday</i>	57	—	44	Gore	2.60
<i>Houston Chronicle</i>	39	Hearst	34	Bush	-1.60
<i>Dallas Morning News</i>	35	—	17	Bush	-0.87
<i>Chicago Sun Times</i>	67	—	58	Bush	-2.70
<i>Boston Globe</i>	72	New York Times	89	Gore	0.50
<i>San Francisco Chronicle</i>	74	Hearst	82	Gore	0.90
<i>Arizona Republic</i>	41	—	20	Bush	-1.00
<i>New York Post</i>	49	—	31	Bush	-1.50
<i>Rocky Mountain News</i>	47	—	28	Bush	-1.30
<i>Denver Post</i>	52	—	35	Gore	3.10
<i>Philadelphia Inquirer</i>	59	Knight Ridder	82	Gore	0.90
<i>Union-Tribune</i>	51	—	34	Bush	-1.60

[†] *USA Today*, *Wall Street Journal*, and *LA Times* are not in this table because those newspapers did not make an endorsement or made a non-endorsement in 2000. *Star Ledger* is not included because the newspaper is not in the sample due to missing ownership information in the ownership data in 2000.

[‡] Missing (—) means that the newspaper is not owned by a group owner. Group owner is defined as a company that owns more than ten daily newspapers in the survey.

(a) Table 4: Influence of top newspapers

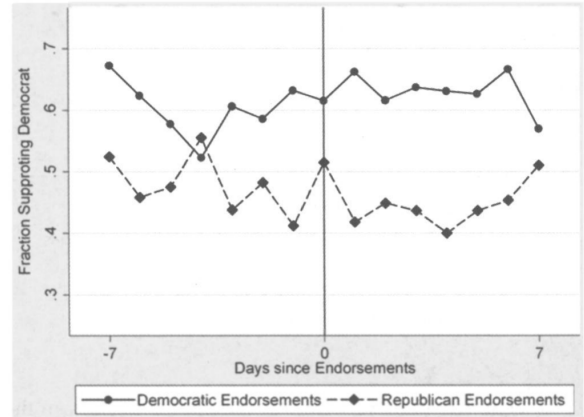


FIGURE 2

(b) Figure 2: Endorsements and voting

6.4 Figures 3 and 4: High- versus low-credibility endorsements

Read:

- Figure 3 restricts the event-study pattern to high-credibility, surprising endorsements.
- The post-endorsement divergence between readers of Democrat-endorsing and Republican-endorsing papers is clearer in this high-credibility sample.
- Figure 4 restricts attention to low-credibility, unsurprising endorsements.
- The effect is much harder to detect for low-credibility endorsements.
- The contrast between Figures 3 and 4 is the visual version of the paper's main argument.

Economic message: Voters respond most when endorsements contradict the newspaper's expected political leaning. This is consistent with rational discounting of media bias.

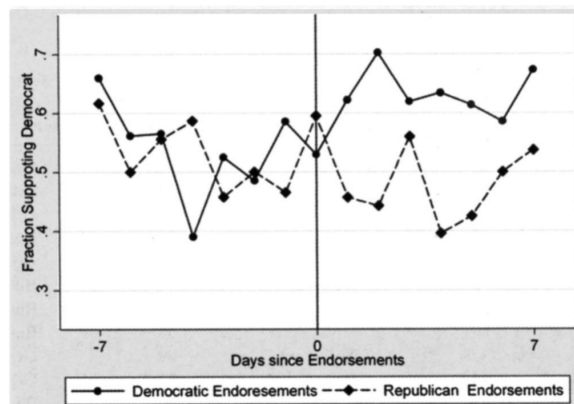


FIGURE 3

High-credibility endorsements and voting

(a) Figure 3: High-credibility endorsements

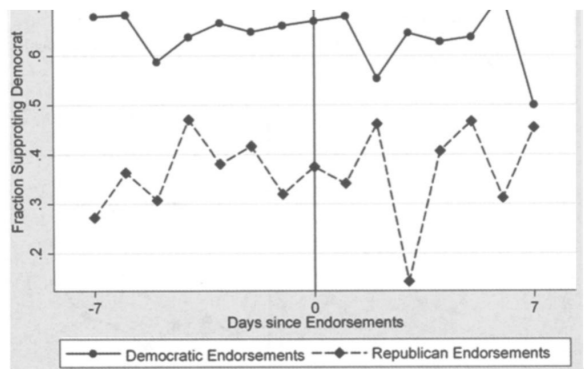


FIGURE 4

Low-credibility endorsements and voting

(b) Figure 4: Low-credibility endorsements

6.5 Tables 5 and 6: Counterfactual effects and alternative explanations

Read:

- Table 5 uses the estimates to simulate counterfactual endorsement scenarios.
- If all newspapers endorsed the Democrat, the Democratic vote share among newspaper readers would rise by about 2.2 percentage points in 2000 and 1.74 in 2004.
- If all newspapers endorsed the Republican, the Democratic vote share among readers would fall by about 2.62 points in 2000 and 3.12 in 2004.
- The implied net difference between all-Democratic and all-Republican endorsements is nearly five percentage points among readers.
- Table 6 addresses alternative explanations, including newspaper-specific trends, ideology-specific trends, credibility-by-date fixed effects, excluding extreme voters, and focusing only on moderates.
- The credibility coefficient remains positive in the main trend and moderate-voter specifications.

Economic message: The estimated effects are electorally meaningful among newspaper readers, and the credibility result is not simply a product of broad trend differences across newspapers or voter ideologies.

TABLE 5
Counterfactual endorsement scenarios

	Year 2000 (%)	Year 2004 (%)
Vote share of the Democratic candidate in sample	53.25	55.41
Predicted change in Democratic vote share <i>among readers</i> if all newspapers made Democratic endorsements	+2.20	+1.74
Predicted change in Democratic vote share <i>among readers</i> if all newspapers made Republican endorsements	-2.62	-3.12
Net effect <i>among readers</i>	+4.82	+4.86
Predicted change in Democratic vote share <i>among voters</i> if all newspapers made Democratic endorsements	+1.64	+1.30
Predicted change in Democratic vote share <i>among voters</i> if all newspapers made Republican endorsements	-1.96	-2.34
Net effect <i>among voters</i>	+3.60	+3.64

(a) Table 5: Counterfactual endorsement scenarios

TABLE 6
Alternative explanations

	Dependent variable: 1 if intend to vote for the Democrat				
	(1)	(2)	(3)	(4)	(5)
After × credibility	0.023* (0.014)	0.024** (0.011)	0.046 (0.031)	0.028** (0.013)	0.051** (0.022)
Newspaper-specific trends	Yes				
Ideology-specific trends		Yes			
Credibility × date fixed effects			Yes		
Voter sample	All	All	All	Exclude extremes	Only moderates
Paper fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	32,014	32,014	32,014	27,905	12,601

Notes: Other control variables are included. Standard errors in parentheses. * denotes 90% significance. ** denotes 95% significance.

(b) Table 6: Alternative explanations

6.6 Tables 7 and 8: Timing and who is influenced

Read:

- Table 7 examines whether newspapers tend to endorse at similar points in the election calendar across 2000 and 2004.
- There is substantial persistence in timing, which weakens the concern that endorsements simply follow high-frequency local campaign events.
- Table 8 studies heterogeneity in responsiveness to endorsements.
- Low-frequency readers are less responsive than high-frequency readers, consistent with an exposure channel.
- Older voters are more responsive in the interaction specification, while born-again Christians are less responsive.
- The paper also emphasizes that moderate voters respond more than ideological extremes.

Economic message: Endorsements influence voters most when voters are exposed to the endorsement and are plausibly open to persuasion. This reinforces the interpretation that endorsements operate through information acquisition.

TABLE 7
Timing of endorsements

	17 Oct, 2004 (%)	24 Oct, 2004 (%)	31 Oct, 2004 (%)	Total (%)
22 Oct, 2000	13.04	10.14	1.45	24.63
29 Oct, 2000	30.43	30.43	7.25	68.11
5 Nov, 2001	1.45	4.35	1.45	7.25
Total	44.92	44.92	10.15	100

Notes: Based upon a sample of 69 newspapers that made endorsements on the three sundays prior to the election in both 2000 and 2004. 5 Nov, 2000 is the Sunday before the 2000 election and 31 Oct, 2004 is the Sunday before the 2004 election.

(a) Table 7: Timing of endorsements

TABLE 8
Who was influenced by newspaper endorsements?

Dependent variable: 1 if intend to vote for the Democrat		
	I	II
After×Credibility	0.045*** (0.018)	-0.182 (0.131)
After×Credibility×Low-frequency readers	-0.032 (0.025)	-0.015 (0.02)
After×Credibility×High school		0.035 (0.066)
After×Credibility×College		0.071 (0.064)
After×Credibility×Male		0.005 (0.023)
After×Credibility×Black		-0.022 (0.032)
After×Credibility×Age		0.009* (0.005)
After×Credibility×Age squared		0.000 (0.000)
After×Credibility×Born-again Christian		-0.063** (0.030)
After×Credibility×Attend religious activities		-0.033 (0.025)
Low-frequency readers	-0.039*** (0.005)	-0.039*** (0.005)
High school	-0.047*** (0.014)	-0.047*** (0.015)
College	-0.013 (0.015)	-0.013 (0.015)
Male	-0.089*** (0.005)	-0.089*** (0.005)
Black	0.441*** (0.008)	0.441*** (0.008)
Age	0.002* (0.001)	0.002* (0.001)
Age squared	-0.000* (0.000)	-0.000* (0.000)
Born-again Christian	-0.149*** (0.006)	-0.149*** (0.006)
Attend religious activities	-0.123*** (0.006)	-0.123*** (0.006)
Income categories	Yes	Yes
Newspaper fixed effect	Yes	Yes
Date and effects	Yes	Yes
Observations	32,014	32,014

Notes: Interaction terms of income categories and credibility are included in the first specification. Standard errors in parentheses, * denotes 90% significance, ** denotes 95% significance, and *** denotes 99% significance.

(b) Table 8: Who was influenced?

6.7 Tables 9 and 10: Alternative credibility measures and robustness

Read:

- Table 9 checks whether the baseline result survives alternative credibility constructions.
- The first alternative uses a surprise measure based on residuals from a linear model of endorsement decisions.
- The historical credibility measures use newspapers' previous endorsement patterns to infer editorial positions.
- The key coefficients are generally positive and are strongest when historical endorsement information is sufficiently rich.
- Table 10 reports additional robustness checks.
- The results allow Democratic and Republican effects to differ, allow effects to differ across 2000 and 2004, use market-level rather than reader-level preferences, add party and ideology controls, and cluster standard errors at the newspaper level.
- The credibility result remains positive and statistically meaningful in the main robustness specifications.

Economic message: The credibility finding is not tied to one exact way of estimating newspaper ideology. Multiple credibility and robustness approaches point in the same direction.

TABLE 9
Alternative credibility measures

Dependent variable: 1 if intend to vote for the Democrat					
	I	II	III	IV	V
After×Surprise measure	0.047** (0.021)				
After×Historical credibility measure		0.027* (0.017)	0.051** (0.024)		
After×Historical surprise measure				0.021 (0.022)	0.129*** (0.043)
Sample	All	Papers with sufficient endorsement history ^a	Papers with more than five historical endorsements	All	Papers with more than five historical endorsements
Paper fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	32,014	14,574	6457	30,446	8793

^a Historical credibility measure is derived from the probability of endorsing the democrat/republican candidates, which is calculated as the fraction of democratic/republican endorsements from historical record. The credibility measure is not well defined when this fraction is either 1 or 0, and newspapers that always supported the same political party are excluded from this specification.

Notes: Other control variables are included. Standard errors in parentheses. * denotes 90 significance, ** denotes 95% significance, and *** denotes 99% significance.

(a) Table 9: Alternative credibility measures

TABLE 10
Additional robustness checks

Dependent variable: 1 if intend to vote for the Democrat					
	I	II	III	IV	V
After×Credibility			0.020 (0.012)	0.022** (0.010)	0.030** (0.014)
After×Credibility-Dem	0.026 (0.024)				
After×Credibility-Rep	-0.033 (0.023)				
After×Credibility-2000		0.015 (0.017)			
After×Credibility-2004		0.047** (0.019)			
Political party categories				Yes	
Ideology categories				Yes	
Standard errors	bootstrap reader	bootstrap reader	bootstrap market	bootstrap reader	cluster reader
First-stage preference	Yes	Yes	Yes	Yes	Yes
Paper fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	32014	32014	32014	31579	32014

Notes: Other control variables are included. Standard errors in parentheses. * denotes 90% significance, ** denotes 95% significance, and *** denotes 99% significance.

(b) Table 10: Additional robustness checks

6.8 Appendix Table B1: Monte Carlo analysis

Read:

- Appendix Table B1 reports Monte Carlo results for the generated-regressor estimator.
- The table varies the number of voters per newspaper and the true endorsement influence parameter β .
- The estimated bias is small in magnitude across cases.
- The mean squared error generally declines as the number of voters per newspaper rises.

Economic message: The Monte Carlo evidence supports the finite-sample reliability of the two-stage credibility estimator. If anything, the simulations suggest that bias works against finding endorsement influence.

APPENDIX TABLE B1
Results from Monte Carlo analysis

#Voters per newspaper	$\beta = 0$		$\beta = 0.1$		$\beta = 0.2$	
	Bias	MSE	Bias	MSE	Bias	MSE
$v = 40$	-0.0044	0.0088	0.0024	0.0114	-0.0028	0.0111
$v = 80$	0.0027	0.0081	0.0008	0.0083	-0.0048	0.0089
$v = 160$	0.0012	0.0055	-0.0006	0.0058	-0.0057	0.0074

Notes: Our bias measure is given by $E(b-\beta)$ where b is our estimator for β . The MSE terms are given by $E(b-\beta)^2$. Further details are provided in Appendix B. MSE, Mean squared error.

Appendix Table B1: Monte Carlo analysis

7 Short summary

- The paper studies newspaper endorsements in the 2000 and 2004 U.S. presidential elections.
- It combines daily voter surveys, newspaper readership, endorsement timing, endorsement choices, reader preferences, and newspaper ownership.

- The theory predicts that voters learn more from endorsements that are surprising given the newspaper’s ideological bias.
- The first stage estimates newspaper ideology using reader preferences and ownership variables.
- The second stage estimates whether credibility-weighted endorsements shift voting intentions after publication.
- The main coefficient on *After* $\times$ *Credibility* is positive and significant.
- A simple endorsement dummy is small and insignificant once credibility is considered.
- High-credibility endorsements have visible and statistically meaningful effects; low-credibility endorsements have little effect.
- Counterfactual simulations imply that endorsements can move vote shares by several percentage points among newspaper readers.
- Endorsements matter more for frequent readers, moderate voters, and older voters; they matter less for born-again Christians in the interaction specification.
- Robustness checks using trends, alternative credibility measures, party and ideology controls, and clustered standard errors support the main interpretation.
- The key takeaway is that voters are influenced by the media but rationally discount predictable bias.

8 Conclusion

8.1 Core conclusion

Chiang and Knight show that newspaper endorsements affect voting intentions, but their influence depends critically on credibility. A predictable endorsement from an ideologically aligned newspaper carries little information. A surprising endorsement from an ideologically distant or neutral newspaper carries more information and produces a larger voting response.

8.2 Economic intuition

The economics is a Bayesian learning problem. Voters care about candidate ideology and candidate quality. They already know newspaper ideology imperfectly through the newspaper’s editorial stance and readership. When a left-leaning paper endorses the Democrat, voters infer that the endorsement may partly reflect bias. When a right-leaning paper endorses the Democrat, voters infer that the newspaper must have received relatively strong favorable information about the Democratic candidate. The same logic holds symmetrically for Republican endorsements.

8.3 Incremental contribution revisited

The paper advances the media-bias literature in three ways. First, it studies influence rather than only slant. Second, it uses exact endorsement timing and daily survey responses to reduce the standard selection problem that readers choose like-minded media. Third, it turns credibility into a measurable object and shows that credibility explains which endorsements move voters.

8.4 Implications for media and democracy

The findings imply that media bias does not mechanically translate into voter persuasion. Biased outlets may be less persuasive precisely because readers discount their predictable endorsements. This does not mean media bias is harmless: biased outlets can still affect information exposure, agenda-setting, and long-run beliefs. But the paper’s evidence suggests that at least in this endorsement setting, voters display sophistication in processing biased information.

8.5 Limitations

The design is observational and cannot completely rule out local shocks correlated with endorsement timing. The credibility measure is generated from an estimated first-stage model, so it depends on assumptions about the mapping from reader preferences and ownership to editorial positions. The analysis studies newspaper readers in presidential elections in 2000 and 2004, so external validity to social media, television, local races, or today’s information environment is not automatic. Finally, voting intention is a survey measure and may not perfectly match actual voting behavior.

8.6 Takeaway

The enduring takeaway is that media influence is strongest when information is credible. Newspapers can move voters, but the persuasive power of an endorsement comes less from the act of endorsing itself and more from whether the endorsement is informative given the source’s expected bias.